

The Geography of Consumption: Local and Sectoral Responses to National Shocks

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Abstract

We develop a Bayesian spatial dynamic factor model to study why common nationwide fluctuations produced sharply different consumer spending responses across U.S. states and sectors after 2020. Using weekly card spending across ten sectors and fifty states, the model decomposes outcomes into a national factor, spatially correlated state-sector exposures, and sectoral dynamics. Exposure to the national factor is strongly spatially clustered, and apparent cross-sector spillovers vanish once this common component is controlled for, reflecting a shared national force rather than genuine transmission. Contact intensity organizes loadings, and national policies affect states through heterogeneous local channels that aggregate data conceal.

Keywords: Bayesian estimation; consumer spending; dynamic factor model; high-frequency data; sectoral dynamics; spatial dependence.

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1 Introduction

Why did the same nationwide forces generate such different spending responses across U.S. states and sectors after 2020? During the pandemic and recovery period, households were exposed to a common sequence of disruptions: lockdowns and reopenings, large fiscal transfers, shifting public-health conditions, inflation, and labor-market volatility. Yet spending did not adjust uniformly. Grocery spending remained comparatively resilient throughout the period, while accommodations and food services, arts and recreation, and transportation experienced much larger early declines and slower normalization. Some states recovered steadily while others experienced prolonged weakness. Understanding these differences requires more than an aggregate time-series perspective. It requires a framework that can explain why common forces were filtered differently across places and categories.

A large empirical literature has documented these patterns using high-frequency transaction data. Chetty et al. (2024) show that the pandemic collapse was concentrated in high-income spending and contact-intensive activity, with sharp heterogeneity across locations. Baker et al. (2020) and Cox et al. (2020) document uneven responses across income groups and spending categories. At the sectoral level, Miescu and Rossi (2021) use a structural VAR identified from financial market news to show that a common COVID-19 uncertainty force depressed Arts, Entertainment and Recreation by 0.87 percent and Transportation by 0.77 percent, while Grocery and Food was the only category to increase. Carvalho et al. (2022) reach similar conclusions using electronic payments data from Portugal, showing that contact-intensive sectors and urban municipalities bore a disproportionate share of the spending collapse. Despite this progress, two important gaps remain. First, existing studies estimate responses at the national or country aggregate level and do not model how a common force transmits differently across geography. Second, they treat sectors as parallel

but independent outcomes and do not ask whether apparent co-movement across categories reflects a shared national driver or genuine lagged cross-sector transmission. Our paper addresses both gaps.

On the fiscal side, Parker et al. (2022) and Baker et al. (2023) show that Economic Impact Payments generated immediate but heterogeneous spending responses across rounds, households, and categories. Andreolli and Surico (2026) further show that spending responses depend on the size of transitory income gains, implying that uniform transfer programs need not produce uniform local outcomes. On the geographic side, Hamann et al. (2023) document that large cities suffered disproportionately during the pandemic in terms of labor market outcomes, consistent with the idea that local economic structure mediates the transmission of common forces. Our earlier work, Katanchian and Parent (2025), established, in a county-level panel without sectoral disaggregation, that geographic spillovers in consumption are substantial and that ignoring spatial structure in factor loadings leads to biased inference, with rural areas displaying stronger responses to fiscal transfers than urban and suburban counties. The present paper takes this line of research further by introducing the sectoral dimension and modeling cross-category dynamics explicitly.

This paper argues that post-2020 spending dynamics were shaped by three forces operating jointly. The first is a common national component that moved many sectors and states at the same time. The second is heterogeneous local exposure to that national component: the same aggregate fluctuation could generate very different state-level responses depending on local economic conditions, policy environments, and sector composition. The third is geography. States are not isolated units, and sensitivity to nationwide forces may be spatially clustered because neighboring states share labor markets, institutions, mobility patterns, and regional economic structures. Our central claim is that uneven spending responses across states and sectors are best understood through this combination of common forces, local exposure, and spatial clustering.

To study this, we develop and estimate a Bayesian spatial dynamic factor model for

weekly state-by-sector spending, measured relative to its pre-pandemic seasonal baseline. The model combines a persistent latent national factor with observed state-week covariates and allows each sector’s state-specific exposure to the common factor to follow a spatial autoregressive structure. A lagged dependence structure across categories is also included to distinguish within-sector persistence from genuine cross-sector transmission. We estimate the model on weekly credit and debit card spending data for ten sectors across the fifty U.S. states from January 2020 through June 2024.

The paper makes four contributions. First, we show that the geography of consumption after 2020 cannot be understood from aggregate trends alone: common forces generated uneven local outcomes because exposure was spatially clustered and sector-specific, connecting the high-frequency transaction literature with the spatial econometrics literature in a unified empirical framework. Second, we provide a Bayesian spatial dynamic factor model for high-frequency state-by-sector panels that jointly captures persistent national dynamics, spatially structured exposure, sectoral persistence, and observed local covariates. Relative to Katanchian and Parent (2025), the key advances are the introduction of the sectoral dimension, the explicit modeling of cross-category dynamics, and the longer sample covering the full recovery through mid-2024. Third, contact intensity emerges as the primary organizing principle behind cross-sector heterogeneity in factor loadings: contact-intensive sectors such as Arts, Entertainment and Recreation and Transportation carry the most negative loadings while goods-oriented sectors carry the largest positive loadings, consistent with the structural evidence in Miescu and Rossi (2021). Fourth, to our knowledge this is among the first papers to show formally, within a state-by-sector spatial dynamic factor setting, that apparent cross-sector co-movement in high-frequency spending data is not credibly different from zero once a latent national factor is included, implying that studies relying on sectoral VARs without controlling for common aggregate forces will attribute to inter-sectoral transmission what is in fact a shared national component. Home improvement spending is positively associated with the first round of Economic Impact Payments and regional price

parity, while accommodations and arts, entertainment and recreation are associated with vaccination progress and labor-market recovery, confirming that uniform national policies transmitted through fundamentally different local mechanisms.

Our findings have direct implications for policy evaluation. Broad national programs such as fiscal transfers or reopening measures may be uniform in design but their associations with local spending are not uniform in practice. Evaluating them through national aggregates alone obscures economically meaningful heterogeneity in local responses. Real-time, state-by-sector monitoring of high-frequency spending data can identify where adjustments lag or where substitution patterns emerge, even when national indicators suggest recovery is on track.

The remainder of the paper is organized as follows. After a brief review of the related literature in Section 2, we develop the theoretical framework in Section 3 and introduce the empirical model and Bayesian estimation procedure in Sections 4 and 5. Section 6 describes the data and covariate set. The main empirical findings appear in Section 7, followed by robustness checks in Section 8 and concluding remarks in Section 9.

2 Literature Review

A large literature uses high-frequency private-sector data to study consumption dynamics during major macroeconomic disruptions. Early work on the COVID-19 period documented sharp spending declines with substantial heterogeneity across households, places, and categories (Baker et al., 2020; Cox et al., 2020; Chetty et al., 2024). Transaction-based datasets made it possible to monitor economic activity in near real time and at fine geographic scales, with Chetty et al. (2024) demonstrating how linked spending and labor-market indicators can track local conditions throughout the pandemic (Alexander and Karger, 2023).

The sectoral dimension of the disruption is well established. Miescu and Rossi (2021) use a daily structural VAR identified by heteroskedasticity in financial markets to show that a

COVID-19 uncertainty shock hit contact-intensive categories far more severely than others: Accommodation and Food Services contracted by 1.07 percent, Arts, Entertainment and Recreation by 0.87 percent, and Transportation by 0.77 percent, while Grocery and Food was the only category to increase. Carvalho et al. (2022) reach similar conclusions using electronic payments data from Portugal, documenting that the collapse was concentrated in contact-intensive sectors and urban municipalities through a combination of composition and behavioral confinement effects. Hamann et al. (2023) find a parallel pattern for regional labor markets in Germany. However, these studies estimate responses at the national or country aggregate level and do not model how a common shock transmits differently across geography, whether sectoral co-movement reflects a common factor or lagged cross-sector spillovers, or how these patterns evolve over a multi-year recovery.

Research on the mechanisms behind the collapse and recovery emphasizes the role of perceived health risk over formal restrictions (Goolsbee and Syverson, 2021; Chetty et al., 2024), motivating frameworks that allow for persistent common forces beyond observed policy variables. The consumption response to fiscal transfers has also received substantial attention: high-frequency evidence shows that Economic Impact Payments generated immediate but heterogeneous spending increases across rounds, households, and categories (Parker et al., 2022; Baker et al., 2023), with responses depending on the size of transitory income gains (Andreolli and Surico, 2026). As excess savings declined, wealth effects and balance-sheet conditions became increasingly relevant (Dao et al., 2024), while vaccination progress supported recovery in contact-intensive sectors (Deb et al., 2022; Cevik, 2026). Across all of these strands, sectors are treated as parallel but independent outcomes: none of these studies models whether cross-sector co-movement reflects a shared national factor or genuine lagged transmission.

Our paper addresses this gap. We develop a Bayesian spatial dynamic factor VARX model that jointly captures persistent national dynamics, spatially structured heterogeneity in exposure to common forces, and sectoral persistence across ten spending categories and

fifty states. A key finding is that apparent cross-sector VAR spillovers arise only when the latent national factor is omitted, suggesting they reflect the autocorrelation of a common force rather than genuine inter-sectoral dynamics.

3 Theoretical motivation

This section develops the economic framework motivating the empirical specification. Three ideas do the main work. First, under standard consumption-smoothing logic, weekly spending growth responds primarily to innovations in expected resources rather than to predictable level movements, which provides a growth-rate theoretical benchmark for the empirical analysis below. Second, sectors differ systematically in their sensitivity to common aggregate forces because they differ in contact intensity: the degree to which consumption requires physical proximity (Mongey et al., 2021). Third, geographic clustering in sensitivity arises because neighboring states share similar economic structures and sector compositions (Krugman, 1991), motivating the spatial autoregressive structure imposed on the factor loadings (LeSage and Pace, 2009). Formal derivations are collected in Appendix A of the Online Appendix.

3.1 Working in growth rates

Consider $N = 50$ U.S. states indexed by i , J spending categories indexed by j , and weeks indexed by t . Households in each state maximize expected lifetime utility over a CES composite of contact-intensive and remote consumption, subject to an intertemporal budget constraint. Log-linearizing the intertemporal Euler equation around a deterministic steady state under $\beta R_f \approx 1$ shows that realized consumption growth reflects innovations to expected lifetime resources rather than predictable level movements (Hall, 1978; Flavin, 1981). This provides a growth-rate theoretical benchmark for weekly sectoral spending, rather than a levels-based one. In practice, weekly spending data exhibit inertia because a fraction of

households consume out of current income rather than smoothing over the life cycle (Campbell and Mankiw, 1989), and because adjustment costs and timing frictions in transaction records generate additional persistence. The estimating equation therefore includes reduced-form lag dynamics, nesting the martingale benchmark when lag coefficients are zero. The empirical outcome variable y_{ijt} , constructed in Section 6 as the seasonally adjusted level of spending relative to its January 2020 baseline following Chetty et al. (2024), is the empirical counterpart available at this frequency; the $\Delta \ln C_{ijt}$ notation of this section therefore serves as a conceptual approximation motivating the sign and dynamics of y_{ijt} , and all estimation is conducted on y_{ijt} directly.

3.2 Contact intensity and sectoral exposure to common forces

Households allocate spending across two types of goods: contact-intensive goods, indexed c , which require physical proximity for consumption, and remote goods, indexed r , which can be consumed without in-person interaction. Following Eichenbaum et al. (2021), the effective price of contact-intensive goods increases with a common aggregate force that captures nationwide health risk, mobility constraints, and aggregate uncertainty. Contact-intensive sectors are more sensitive to this force because their consumption requires physical proximity, which becomes costly when aggregate conditions deteriorate. Remote and goods-oriented sectors are less sensitive, and for categories where home consumption substitutes directly for in-person activity, such as grocery and home improvement, sensitivity may be negative as households substitute toward at-home spending (Guerrieri et al., 2022).

We capture the common force through a latent factor f_t that represents the shared component of sectoral spending, relative to baseline, across states, absorbing persistent nationwide movements in aggregate demand, health risk, and uncertainty that are not spanned by observed covariates. Rather than interpreting f_t as the level of an underlying structural force, we follow standard practice in latent factor models and treat it directly as the common latent component of spending relative to baseline (Stock and Watson, 2002; Bai and

Ng, 2002). The contact-intensity framework motivates why this common force transmits with different intensity and direction across sectors: categories that require physical proximity load negatively on f_t during periods of elevated aggregate stress, while goods-oriented categories load positively as households substitute toward at-home consumption.

Combining the intratemporal optimality condition with the intertemporal Euler equation and log-linearizing around a deterministic steady state (see Appendix A.3 of the Online Appendix for the full derivation), sectoral spending growth satisfies:

$$\Delta \ln C_{ijt} \approx \mu_{ij} + b'_{ij}x_{it} + A_{ij}f_t + \varepsilon_{ijt}, \quad (1)$$

where x_{it} collects observed state-week covariates capturing labor-market conditions, policy timing, fiscal transfers, and cost-of-living measures, and $A_{ij} = -\sigma\tilde{\lambda}_j$ is the factor loading for sector j in state i , with $\tilde{\lambda}_j = \lambda_j - \lambda_r$ measuring the excess contact intensity of sector j relative to the remote goods baseline and $\sigma > 0$ the intratemporal elasticity of substitution. In the main text we maintain the restriction $\sigma = 1/\theta$, where θ is the intertemporal curvature of utility, so that:

$$A_{ij} = -\sigma\tilde{\lambda}_j = -\frac{\tilde{\lambda}_j}{\theta}. \quad (2)$$

The theoretical framework motivates the sign and sectoral heterogeneity of the factor loadings through the contact-intensity mechanism. The coefficients on observed covariates b_{ij} are treated as reduced-form controls rather than structural elasticities, and we make no claim that they correspond to structural parameters derived from the household problem.

Equation (2) delivers the following predictions. Contact-intensive sectors have large positive $\tilde{\lambda}_j$ and therefore negative loadings, while the reference remote-goods type has $\tilde{\lambda}_r = 0$ by normalization. Positive estimated loadings for goods-oriented sectors arise whenever a sector's own contact intensity falls below the reference type, $\lambda_j < \lambda_r$, in particular when $\lambda_r < 0$ itself, meaning remote consumption becomes effectively cheaper when aggregate conditions worsen (Guerrieri et al., 2022). The framework also predicts goods-services substitution when

aggregate conditions worsen (Guerrieri et al., 2022): households shift toward remote and goods categories, so contact-intensive and goods-oriented sectors should move in opposite directions in response to the common force, consistent with the pandemic-era reallocation documented by Alexander and Karger (2023).

These predictions are confirmed by the structural evidence in Miescu and Rossi (2021), who find that Arts, Entertainment and Recreation contracted by 0.87 percent and Transportation by 0.77 percent in response to a common COVID-19 uncertainty force, while Grocery and Food was the only expenditure category to increase. Our estimated loadings replicate this pattern: Arts, Entertainment and Recreation and Transportation carry the most negative mean loadings, while General Merchandise and Sport, Hobby, Books, and Music carry the largest positive loadings.

Health Care is a likely exception to this framework since health care spending is shaped primarily by state-level insurance structures, Medicaid expansion status, and provider availability rather than by the physical proximity required for consumption. As we show in Section 7.1, this sector exhibits weaker spatial clustering and a lower share of significant loadings than any other category, consistent with institutional heterogeneity playing a larger role than contact intensity in determining local exposure to national forces.

3.3 Geographic clustering in sectoral exposure

Equation (2) explains why loadings vary across sectors through $\tilde{\lambda}_j$ but does not by itself explain cross-state variation within the same sector. Indeed, under the restriction $\sigma = 1/\theta$ maintained throughout the paper, the theoretical loading $A_{ij} = -\tilde{\lambda}_j/\theta$ carries no state index on its right-hand side at all: theory pins down only the sectoral component of exposure to the common factor. Under this restriction, the structural model does not itself generate cross-state variation in A_{ij} . We therefore introduce state-level heterogeneity in loadings as a reduced-form spatial extension, motivated empirically by differences across states in local economic structure, sector composition, and consumption patterns – captured through ϕ_i ,

the preference weight on contact-intensive consumption, which is not itself derived from equation (2) once $\sigma = 1/\theta$ is imposed. Because ϕ_i reflects industrial structure, it is spatially correlated across neighboring states through shared industrial structures and labor markets (Krugman, 1991; Blanchard and Katz, 1992), motivating the spatially clustered pattern captured by the SAR structure below. This provides the economic motivation for the spatial autoregressive structure imposed on A_j , which we treat as a reduced-form device capturing the consequences of spatially correlated economic structure rather than a direct implication of the household first-order conditions: the sectoral component of exposure is structural, while its state-specific dispersion is a reduced-form spatial extension layered on top.

Formally, for each sector j , stack state loadings into $A_j = (a_{1j}, \dots, a_{Nj})'$ and model their spatial dependence as:

$$A_j = (I_N - \rho_j W)^{-1} L_j, \quad L_j \sim \mathcal{N}(0, \sigma_\eta^2 I_N), \quad (3)$$

where W is a row-normalized spatial weights matrix based on four-nearest-neighbor state centroid distances (LeSage and Pace, 2009), ρ_j governs the strength of spatial dependence in sector j and lies within the admissible interval for which $(I_N - \rho_j W)$ is nonsingular, determined by the eigenvalues of W (Section 4 gives the exact bounds), and σ_η^2 is a single innovation variance shared across sectors, estimated jointly using information pooled across all J sectors. The parameter ρ_j is sector-specific because the degree of regional economic integration differs across categories: sectors tied to local physical infrastructure cluster differently from those linked to regional tourism or transportation networks.

The full empirical specification combines (1) and (3) with a VAR component for within- and cross-sector lag dynamics and observed state-week covariates. The complete model is introduced in Section 4.

4 Empirical specification

Section 3 motivates modeling weekly sectoral spending, measured relative to its pre-pandemic seasonal baseline, as a linear function of a persistent latent national factor, observed state-week covariates, and reduced-form lag dynamics. This section translates that motivation into the estimating equation used throughout the paper. The data are described in Section 6 and the likelihood, priors, and MCMC algorithm are presented in Section 5.

The panel covers $N = 50$ U.S. states indexed by i , $J = 10$ spending sectors indexed by j , and $T = 231$ weeks indexed by t . The outcome variable y_{ijt} is the weekly, seasonally adjusted level of spending relative to its January 2020 baseline, constructed from the Opportunity Insights Economic Tracker and defined formally in Section 6. For each state-week, sectoral outcomes are collected into the $J \times 1$ vector $y_{it} = (y_{i1t}, \dots, y_{iJt})'$, and x_{it} denotes the $K \times 1$ vector of observed state-week covariates described in Section 6.

The baseline specification models sectoral spending, relative to baseline, as:

$$y_{it} = \mu + \Gamma y_{i,t-1} + Bx_{it} + A_i f_t + \varepsilon_{it}, \quad \varepsilon_{it} \sim \mathcal{N}(0, \Sigma_{\varepsilon,i}), \quad (4)$$

where μ is a $J \times 1$ vector of sector intercepts, Γ is a $J \times J$ VAR(1) coefficient matrix capturing within- and cross-sector lag dependence, B is a $J \times K$ matrix of responses to observed covariates, f_t is the scalar latent national factor ($q = 1$), A_i is the $J \times 1$ state-specific loading vector governing the transmission of f_t to sectoral outcomes in state i , and $\Sigma_{\varepsilon,i} = \text{diag}(\sigma_{\varepsilon,i1}^2, \dots, \sigma_{\varepsilon,iJ}^2)$ is state i 's $J \times J$ diagonal block of the full $NJ \times NJ$ idiosyncratic covariance matrix Σ_{ε}^* introduced in Section 5.1; idiosyncratic variances are allowed to differ across both states and sectors, not just across sectors. The diagonal structure of $\Sigma_{\varepsilon,i}$ assumes that disturbances are uncorrelated across sectors conditional on the common factor, a standard tractability assumption in large factor models (Bai and Ng, 2002). Relative to the theoretical sectoral growth equation of Section 3, which allows μ_{ij} and b_{ij} to vary by state, the empirical specification in equation (4) imposes $\mu_{ij} = \mu_j$ and $b_{ij} = b_j$ (equivalently, the

rows of B) pooled across states within each sector; state-level heterogeneity instead enters through the state-specific loadings A_i and the state-varying covariates x_{it} themselves.

With a single common factor, f_t is a scalar and A_i reduces to a $J \times 1$ loading vector for each state. The key structural difference relative to Katanchian and Parent (2025) is how the cross-sectional dimension of factor exposure is handled. In that paper, each location is assigned a p -dimensional loading vector $\Lambda_i = (\lambda_{i,1}, \dots, \lambda_{i,p})'$ on a p -dimensional factor F_t , with the multiple factors jointly capturing common variation across a single aggregate spending index at the county level. In the present paper, the multi-factor structure is replaced by a multi-sector structure: rather than allowing one location to load differently on multiple factors, we allow one factor to transmit differently to multiple sectors within the same state. Each state i receives a distinct loading a_{ij} for each sector $j = 1, \dots, J$, so that the same national force f_t can generate a large positive response in General Merchandise and a large negative response in Transportation within the same state, depending on the sector's contact intensity and economic structure. The J sector-specific loadings collected in the $J \times 1$ vector A_i therefore play the role that the p -dimensional factor vector played in the earlier paper: they are the primary vehicle for capturing heterogeneity in how the common national force transmits to different parts of the economy. The spatial autoregressive structure described below is applied sector-by-sector, with a separate parameter ρ_j governing the geographic clustering of exposure in each category, rather than a single spatial parameter applied to a vector of factor loadings as in Katanchian and Parent (2025).

The full stacked loading vector across all states and sectors is the $NJ \times 1$ vector $A = (A'_1, \dots, A'_N)'$, and for each sector j the $N \times 1$ vector of state loadings $A_j = (a_{1j}, \dots, a_{Nj})'$ follows a spatial autoregressive process. For notational convenience, define the stacked $NJ \times 1$ outcome vector $y_t \equiv (y'_{1t}, \dots, y'_{Nt})'$. The common component in the stacked system is then Af_t . Section 5 formalizes the stacked representation used in the likelihood and MCMC algorithm.

The $J \times J$ matrix Γ plays a central role in distinguishing genuine cross-sector dynam-

ics from spurious co-movement driven by a shared national force. The diagonal elements Γ_{jj} capture own-sector persistence arising from habit formation, slow reallocation, and sector-specific adjustment costs (Fuhrer, 2000; Carroll, 2000), though part of this persistence is mechanical: because y_{ijt} is itself a cumulative level relative to a fixed baseline rather than an independently drawn period observation, Γ_{jj} also absorbs some of the index’s own construction-induced persistence and should not be interpreted purely as an economic behavioral parameter. The off-diagonal elements Γ_{jk} , $j \neq k$, capture lagged cross-sector spillovers. A central finding of the paper, documented in Section 7 and examined in Section 8, is that these off-diagonal elements are not credibly different from zero once the latent factor and spatial exposure are controlled for. Apparent cross-sector co-movement in spending data therefore reflects a shared national force rather than genuine lagged transmission across categories.

The scalar factor f_t captures persistent common movements in spending, relative to baseline, that are not explained by observed covariates. It follows a stationary AR(1) process:

$$f_t = \phi f_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0, \sigma_u^2), \quad |\phi| < 1, \quad (5)$$

with initial condition $f_1 \sim \mathcal{N}(0, C_1)$. The stationarity restriction is imposed through a truncated normal prior as described in Section 5. The estimated posterior mean of ϕ is 0.98, indicating that the national factor moves slowly relative to the weekly sampling frequency. As with the diagonal elements of Γ , this near-unit-root persistence should not be read purely as evidence of a slow-moving economic condition: because y_{ijt} is itself a cumulative level relative to a fixed baseline rather than a period-to-period draw, any common factor extracted from it will tend to inherit substantial persistence mechanically, independent of how persistent the underlying economic conditions actually are. This slow-moving character is nonetheless consistent with the interpretation of f_t as a broad national condition encompassing health risk, aggregate uncertainty, and demand conditions rather than high-frequency noise (Miescu

and Rossi, 2021). Because the prior on ϕ is truncated to the stationary region $|\phi| < 1$ (Section 5.3), posterior inference for ϕ is conducted entirely over stationary AR(1) processes and does not rely on the nonstandard sampling theory that complicates classical inference near a unit root (Sims and Uhlig, 1991).

Heterogeneous state-level exposure to this common force is captured through the loading vectors A_i . Rather than treating these loadings as free parameters for each state, we impose a spatial autoregressive structure that allows neighboring states to share similar sensitivity to the national factor within each sector. For each sector j , the $N \times 1$ vector of state loadings $A_j = (a_{1j}, \dots, a_{Nj})'$ follows:

$$A_j = (I_N - \rho_j W)^{-1} L_j, \quad L_j \sim \mathcal{N}(0, \sigma_\eta^2 I_N), \quad (6)$$

where W is a row-normalized $N \times N$ spatial weights matrix based on four-nearest-neighbor state centroid distances, ρ_j governs the strength of spatial dependence in sector j , and σ_η^2 is a single innovation variance shared across sectors, updated using information pooled across all J sectors rather than estimated separately for each. Because the four-nearest-neighbor construction need not be symmetric even before row-normalization (state k being among state i 's nearest neighbors does not imply the reverse), W is not guaranteed to be symmetric, and its eigenvalues need not all be real. The admissible values of ρ_j are those for which $(I_N - \rho_j W)$ is nonsingular (LeSage and Pace, 2009). By the Perron–Frobenius theorem, the dominant eigenvalue of a row-normalized nonnegative matrix is real and equal to 1 regardless of symmetry, giving an upper bound of 1. The lower bound is $1/\text{Re}(\lambda_{\min})$, where $\lambda_{\min}$ is the eigenvalue of W with the smallest (most negative) real part, computed numerically rather than assumed to equal -1 . Estimation is conducted over this full admissible interval, with no prior restriction to positive values; the posterior mean of ρ_j is positive in every sector (Table 3), indicating that neighboring states are estimated to share similar sensitivity to the national factor within sector j , generating geographic clustering in how aggregate conditions

transmit to local sectoral spending. The parameter ρ_j is sector-specific because the degree of regional economic integration differs across categories, as discussed in Section 3.3.

The common component $A_i f_t$ is identified only up to a scale and sign normalization. We fix the scale by setting the loading for Alabama in the Apparel and Accessories sector to unity and impose a positive sign convention so that f_t is interpretable as a pro-cyclical national spending force: a positive realization corresponds to above-trend spending in sectors with positive loadings. The normalization is applied at each MCMC iteration as described in Section 5. This choice affects only the scale and sign of f_t and the absolute magnitude of the loadings; all relative loadings across states and sectors, spatial dependence parameters ρ_j , VAR coefficients, and covariate responses are invariant to it. We verified that alternative normalizations produce factor paths and loading maps that differ only by a scalar multiple, leaving all substantive conclusions unchanged.

Taken together, the model separates three channels through which spending, relative to baseline, co-moves across states and sectors. The latent national factor f_t generates contemporaneous co-movement through the sector-specific loading structure A_i , which replaces the multi-factor structure of Katanchian and Parent (2025) with a richer sectoral transmission mechanism. Geographic clustering in factor sensitivity, captured by the sector-specific parameters ρ_j , generates regional patterns in how that national force transmits locally within each spending category. Within-sector lag persistence, captured by the diagonal elements of Γ , accounts for short-run inertia in each category. A key empirical result is that the first two channels account for essentially all observed co-movement: once f_t and the spatial loadings are controlled for, the off-diagonal elements of Γ are not credibly different from zero and own-sector persistence remains the dominant dynamic feature of the data.

5 Estimation procedure

We present the estimation procedure for the baseline specification introduced in Section 4. The latent factor f_t is a scalar, the loading vector A is $NJ \times 1$, and the factor path $F = (f_1, \dots, f_T)'$ is $T \times 1$. The extension to a multi-dimensional factor is developed in Katanchian and Parent (2025), which establishes the computational properties of the banded Cholesky simulation smoother for arbitrary factor dimension.

5.1 Stacked representation

Building on the specification in Section 4, we develop the stacked representation used in the likelihood and MCMC algorithm. Idiosyncratic variances are allowed to differ across all NJ state-sector equations while disturbances remain conditionally independent across those equations given the latent factor. Let

$$\Sigma_\varepsilon^* = \text{diag}(\sigma_{\varepsilon,1}^2, \dots, \sigma_{\varepsilon,NJ}^2), \quad (7)$$

where the NJ diagonal elements index all state-sector pairs.

The $NJ \times 1$ stacked outcome vector at time t is $y_t \equiv (y'_{1t}, \dots, y'_{Nt})'$, where each y_{it} is $J \times 1$. The full loading vector $A \equiv (A'_1, \dots, A'_N)'$ is $NJ \times 1$, where each A_i is the $J \times 1$ vector that collects the J sector-specific loadings for state i . For each sector j , the $N \times 1$ vector $A_j = (a_{1j}, \dots, a_{Nj})'$ collects the loading of sector j across all N states and follows the SAR process in equation (6). Stacking these J sector-level vectors yields the full $NJ \times 1$ loading vector A .

Let $\beta \equiv \text{vec}(B)$, which is $JK \times 1$, and let X_t denote the stacked $NJ \times JK$ design matrix for observed covariates, constructed so that $X_t\beta$ reproduces the term Bx_{it} across all states. The stacked measurement equation at time t is:

$$y_t = (\mathbf{1}_N \otimes \mu) + (I_N \otimes \Gamma)y_{t-1} + X_t\beta + Af_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma_\varepsilon^*), \quad (8)$$

where $(\mathbf{1}_N \otimes \mu)$ is $NJ \times 1$ and repeats the sector intercept vector across states, and $(I_N \otimes \Gamma)$ is $NJ \times NJ$ and applies the same $J \times J$ VAR matrix to each state's sectoral vector. Note that Af_t is $NJ \times 1$ since A is $NJ \times 1$ and f_t is a scalar.

The likelihood is conditioned on the first observed week rather than on a genuine pre-sample period: throughout this subsection T denotes the number of one-week-ahead transitions actually used in estimation, equal to $T_{\text{eff}} = 230$ in the notation introduced in Section 6.2 below, and y_0 below denotes the first observed week of the raw 231-week panel rather than an unobserved value prior to the sample. Stacking over $t = 1, \dots, T$ (in this re-indexed sense) yields:

$$y \equiv (y'_1, \dots, y'_T)', \quad y_- \equiv (y'_0, \dots, y'_{T-1})', \quad X \equiv (X'_1, \dots, X'_T)', \quad F \equiv (f_1, \dots, f_T)', \quad (9)$$

where y and y_- are $TNJ \times 1$, X is $TNJ \times JK$, and F is $T \times 1$. The full stacked system is:

$$y = (\mathbf{1}_T \otimes \mathbf{1}_N \otimes \mu) + (I_T \otimes (I_N \otimes \Gamma))y_- + X\beta + (F \otimes A) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I_T \otimes \Sigma_\varepsilon^*), \quad (10)$$

where the common component $(F \otimes A)$ is $TNJ \times 1$: at each time t , the scalar f_t multiplies the $NJ \times 1$ vector A to produce the common component $f_t A$, and stacking these over t yields $(F \otimes A) = (f_1 A', \dots, f_T A')'$.

5.2 Likelihood

Conditional on $(\mu, \Gamma, \beta, F, A, \Sigma_\varepsilon^*)$, the likelihood implied by (10) is Gaussian:

$$p(y \mid \mu, \Gamma, \beta, F, A, \Sigma_\varepsilon^*, X) = (2\pi)^{-NJT/2} |I_T \otimes \Sigma_\varepsilon^*|^{-1/2} \exp \left\{ -\frac{1}{2} e' (I_T \otimes \Sigma_\varepsilon^*)^{-1} e \right\}, \quad (11)$$

where

$$e \equiv y - (\mathbf{1}_T \otimes \mathbf{1}_N \otimes \mu) - (I_T \otimes (I_N \otimes \Gamma))y_- - X\beta - (F \otimes A) \quad (12)$$

is the $TNJ \times 1$ residual vector. The factor persistence parameter ϕ , the innovation variance σ_u^2 , and the spatial parameters $\{\rho_j\}$ enter through the prior and state equations rather than directly through the Gaussian likelihood.

5.3 Prior distributions

We assign conditionally conjugate priors to regression and variance components. Gaussian priors are set for the sector intercepts, covariate coefficients, and VAR coefficients:

$$\mu \sim \mathcal{N}(c_\mu, T_\mu), \quad \beta \sim \mathcal{N}(c_\beta, T_\beta), \quad \text{vec}(\Gamma) \sim \mathcal{N}(c_\Gamma, T_\Gamma), \quad (13)$$

with $c_\mu = 0$, $c_\beta = 0$, and $c_\Gamma = 0$. The prior for Γ places stronger shrinkage on off-diagonal elements than on diagonal elements. This asymmetry reflects two considerations. First, in the presence of a latent common factor and observed covariates, direct lagged transmission from one spending category to another within a week is unlikely to be economically large, motivating tighter shrinkage on off-diagonal elements. Second, with $J = 10$ sectors the unrestricted off-diagonal block contains 90 free parameters, and regularization through an informative prior reduces the risk of overfitting in finite samples. Diagonal elements are therefore assigned prior variance $\tau_{\text{diag}}^2 = 0.25$ while off-diagonal elements are assigned $\tau_{\text{off}}^2 = 4 \times 10^{-4}$, corresponding to prior standard deviations of 0.50 and 0.02 respectively.¹ The finding that cross-sector spillovers are not credibly different from zero is not a prior artifact: Section 8 shows that when the common factor is removed and equal prior variances are used for all elements of Γ , significant cross-sector co-movement emerges, confirming that its absence in the baseline reflects the presence of the latent factor rather than the tightness of the prior. To ensure stability of the VAR dynamics, any draw of Γ whose spectral radius exceeds 0.99 is rejected and redrawn from the same conditional posterior, so that all retained draws satisfy $\rho(\Gamma) \leq 0.99$. Because this is standard rejection sampling, retained draws are

¹Results under alternative off-diagonal prior variances $\tau_{\text{off}}^2 \in \{0.01, 0.04, 0.25\}$ are available upon request.

exact draws from the Gaussian conditional posterior truncated to the stability region, with no distortion of the underlying distribution. For each of the NJ state-sector equations the idiosyncratic variance follows an inverse-gamma prior:

$$\sigma_{\varepsilon,h}^2 \sim IG(n_{\varepsilon,h}/2, s_{\varepsilon,h}/2), \quad h = 1, \dots, NJ. \quad (14)$$

The latent factor follows the AR(1) process introduced in Section 4, with stationarity imposed through a truncated normal prior and an inverse-gamma prior on the innovation variance:

$$\phi \sim \mathcal{N}_{tr_{(-1,1)}}(m_\phi, S_\phi), \quad \sigma_u^2 \sim IG(n_u/2, s_u/2). \quad (15)$$

5.4 Simulation smoother for the factor path

Let $F = (f_1, \dots, f_T)'$ denote the $T \times 1$ factor path. The AR(1) prior implies the linear system $HF = \xi$, where H is the $T \times T$ bidiagonal matrix:

$$H_{t,t} = 1, \quad H_{t,t-1} = -\phi, \quad t \geq 2, \quad (16)$$

and zero elsewhere. The innovation vector $\xi = (\xi_1, \dots, \xi_T)'$ has $\xi_1 \sim \mathcal{N}(0, \sigma_u^2/(1 - \phi^2))$ and $\xi_t \sim \mathcal{N}(0, \sigma_u^2)$ for $t \geq 2$. With $S = \text{diag}(\sigma_u^2/(1 - \phi^2), \sigma_u^2, \dots, \sigma_u^2)$, the prior precision of F is:

$$M = H'S^{-1}H, \quad (17)$$

which is $T \times T$, tridiagonal, and sparse.

Conditional on $(y, \mu, \Gamma, \beta, A, \Sigma_\varepsilon^*, \phi, \sigma_u^2)$, the posterior for F is Gaussian with precision

matrix Ω and posterior mean $\hat{F}$:

$$\Omega = M + \left(A'(\Sigma_\varepsilon^*)^{-1} A \right) I_T, \quad (18)$$

$$b = \left(A'(\Sigma_\varepsilon^*)^{-1} \right) (y_t - (\mathbf{1}_N \otimes \mu) - (I_N \otimes \Gamma)y_{t-1} - X_t\beta) \text{ stacked over } t, \quad (19)$$

$$\hat{F} = \Omega^{-1}b. \quad (20)$$

Since A is $NJ \times 1$ and Σ_ε^* is diagonal, $A'(\Sigma_\varepsilon^*)^{-1}A$ reduces to the positive scalar $\sum_{h=1}^{NJ} a_h^2/\sigma_{\varepsilon,h}^2$, which we denote κ . The posterior precision is therefore $\Omega = M + \kappa I_T$, which is $T \times T$ and tridiagonal since M is tridiagonal and κI_T is diagonal. This structure means the Cholesky factorization of Ω costs $O(T)$ rather than $O(T^3)$.

An exact draw of F is obtained via the banded Cholesky simulation smoother, factoring $\Omega = CC'$ with C lower bidiagonal; the explicit forward/backward substitution and sampling steps are given in Appendix C.3 of the Online Appendix. This delivers an exact draw from the posterior at cost $O(T)$ per iteration (McCausland et al., 2011; Katanchian and Parent, 2025).

5.5 Spatial loadings and posterior update

The SAR structure on the loadings follows equation (6) of Section 4. For each sector j , the $N \times 1$ vector $A_j = (a_{1j}, \dots, a_{Nj})'$ of state loadings follows:

$$A_j = (I_N - \rho_j W)^{-1} L_j, \quad L_j \sim \mathcal{N}(0, \sigma_\eta^2 I_N), \quad (21)$$

where σ_η^2 is a single innovation variance shared across sectors. The full $NJ \times 1$ loading vector A is assembled from the J sector-level vectors A_j by interleaving them according to the state-major ordering used throughout the paper: the $((i-1)J + j)$ -th element of A equals a_{ij} , the i -th element of A_j , so that $A = (A'_1, \dots, A'_N)'$ with each A_i collecting the J sector-specific loadings for state i , consistent with the stacking convention introduced in

Section 5.1.

Conditional on $(F, \mu, \Gamma, \beta, \Sigma_\varepsilon^*, \{\rho_j\}, \sigma_\eta^2)$, the loadings A_j follow the SAR structure in equation (6), with L_j the underlying spatial innovation. The spatial dependence parameters ρ_j are updated using a griddy Gibbs step based on log-determinant interpolation over the admissible interval implied by the spectral bounds of W (LeSage and Pace, 2009). The shared innovation variance σ_η^2 is updated from its inverse-gamma full conditional, pooling information across all J sectors. Full conditional expressions and implementation details are reported in Appendix C of the Online Appendix.

5.6 Posterior inference

Posterior inference is based on 100,000 MCMC iterations with the first 1,000 discarded as burn-in, yielding 99,000 retained draws. Each iteration cycles through the following blocks in sequence: regression coefficients (μ, β) via Gaussian conjugate updates, with the factor path integrated out analytically rather than conditioned on; spatial factor loadings A via a Gaussian conjugate update, with the SAR structure entering as the prior precision; the full factor path F via the tridiagonal Cholesky simulation smoother described in Section 5.4; idiosyncratic variances Σ_ε^* via inverse-gamma updates; factor persistence ϕ via a Metropolis-Hastings step and innovation variance σ_u^2 via an inverse-gamma update; the shared spatial innovation variance σ_η^2 via an inverse-gamma update; spatial dependence parameters $\{\rho_j\}$ via griddy Gibbs; and VAR coefficients Γ via Gaussian conjugate updates subject to rejection sampling under the spectral radius restriction. Convergence is assessed using trace plots and by verifying that posterior summaries are stable across subsamples of the retained draws.

6 Data

6.1 Dependent variable

The outcome variable is constructed from the Opportunity Insights Economic Tracker, available at <https://www.tracktherecovery.org/>. The underlying transaction data come from Affinity Solutions Inc., which aggregates anonymized credit and debit card records to support bank loyalty programs and related financial services. These records cover roughly one in ten dollars of card-based spending in the United States and are available at the state, sector, and week level beginning in January 2019. Because spending is assigned to the cardholder’s home billing address rather than the point of purchase, the data capture where consumers live rather than where they shop. The series is representative of card-based spending but not of total household expenditure, since cash transactions and spending categories rarely paid for by card are not covered (Chetty et al., 2024).

The Opportunity Insights series is not a raw spending measure but an already-constructed index, released publicly through two adjustments applied by Chetty et al. (2024) before the data ever reach researchers. First, to remove seasonal patterns, each week’s spending in 2020 onward is expressed relative to the corresponding week in 2019. Second, the resulting seasonally adjusted series is indexed to the average level of the first four complete weeks of January 2020, so that the publicly released series reports a percentage deviation from that baseline rather than a raw spending level – a design choice made partly for the confidentiality reasons described in Chetty et al. (2024). We do not observe or process raw spending S_{ijt} ourselves; we take the published series directly, aggregating to the weekly level by averaging observations within each calendar week for the period reported at the daily frequency, and using the series as released at the weekly frequency from June 2022 onward. Denoting the underlying raw spending in state i , sector j , and week t as S_{ijt} for exposition, the published

outcome variable can be written as:

$$y_{ijt} \equiv \frac{S_{ijt}/S_{ij,\text{Jan2020}}}{S_{ij,2019(t)}/S_{ij,\text{Jan2019}}} - 1, \quad (22)$$

where $S_{ij,2019(t)}$ is spending in the corresponding week of 2019, and $S_{ij,\text{Jan2020}}$ and $S_{ij,\text{Jan2019}}$ denote average spending over the first four complete weeks of January 2020 and January 2019, respectively. The numerator indexes current spending to its own-year January 2020 baseline; the denominator applies the identical indexing to the corresponding week one year earlier, using the January 2019 baseline. Dividing the two removes seasonal patterns common to both years, and subtracting one converts the resulting ratio into a proportional deviation: y_{ijt} is the amount by which the year-over-year change in spending at week t exceeds (or falls short of) the same year-over-year change evaluated during the January 2020 reference period. Because both $S_{ij,\text{Jan2020}}$ and $S_{ij,\text{Jan2019}}$ are fixed reference points rather than moving one-period-back comparisons, y_{ijt} is a seasonally adjusted *cumulative* proportional deviation from the January 2020 baseline, not a week-to-week growth rate. A positive value of y_{ijt} indicates that spending in week t runs above its pre-pandemic seasonal path by more than it did, on average, during the January 2020 baseline period; a negative value indicates spending running below that path. This construction follows the publicly released Opportunity Insights series exactly, indexed to January 2020 by Chetty et al. (2024) as described above.

As discussed in Section 3, the theoretical object motivated by consumption-smoothing logic is a period-to-period log growth rate, $\Delta \ln C_{ijt}$, while y_{ijt} is a proportional, non-log cumulative deviation from a fixed baseline. We treat $\Delta \ln C_{ijt}$ as a conceptual motivation for the sign and persistence properties of y_{ijt} rather than as a literal description of the estimating equation, and conduct all estimation on y_{ijt} directly, following the same approach used in Katanchian and Parent (2025). For each state-week, sectoral outcomes are collected into the $J \times 1$ vector $y_{it} = (y_{i1t}, \dots, y_{iJt})'$.

6.2 Sample and sectors

The sample covers January 2020 through June 2024, yielding $T = 231$ weekly observations per state-sector series. Inclusion of one lag of y_{it} in the VAR(1) component of the baseline specification means this component uses $t = 2, \dots, T$, i.e. $T_{\text{eff}} = 230$ lagged transitions per state; the initial observation is retained in estimation of the contemporaneous factor and covariate components. The panel contains $N = 50$ U.S. states, excluding the District of Columbia, giving $N \times T_{\text{eff}} = 11,500$ state-week observations per sector entering the VAR component.

Spending, relative to baseline, is observed in $J = 10$ sectors: Apparel and Accessories; Accommodations and Food Services; Arts, Entertainment, and Recreation; Auto and Gas; Health Care; Home Improvement; General Merchandise; Grocery and Food; Sport, Hobby, Books, and Music; and Transportation. These ten categories span both contact-intensive services and remote goods, covering the full range of sectoral heterogeneity that the model is designed to capture.

Figure 1 plots the national average weekly spending, relative to its January 2020 baseline, for each sector over the full sample period. The figure makes the central empirical puzzle of the paper immediately visible. Transportation remained persistently negative for nearly two years before recovering, while General Merchandise and Home Improvement surged during the same period and remained well above their pre-pandemic baseline. Arts, Entertainment, and Recreation collapsed sharply in early 2020 and recovered slowly and unevenly. Grocery and Food barely dipped in the acute phase and remained positive throughout, consistent with substitution toward at-home consumption. The same nationwide disruption therefore generated markedly different trajectories across categories, with contact-intensive sectors and goods-oriented sectors moving in opposite directions for an extended period. This sectoral divergence motivates the model’s central feature: a single common national factor that transmits with different intensity and direction across sectors through the loading vectors A_i .

6.3 Covariates

The covariate vector x_{it} collects observed state-week fundamentals that enter the model as controls for local economic conditions and policy environment. Labor-market conditions are measured by initial unemployment claims, employment growth, and lagged personal income. Policy timing is captured by indicators for the first date of nonessential business closures and stay-at-home orders in each state. Fiscal transfers are represented by the three rounds of Economic Impact Payments (EIP1–EIP3), constructed from IRS administrative data on disbursements and aligned to the weekly panel, with non-disbursement weeks coded as zero. Public-health progress is captured by vaccination rates. Cost-of-living and price conditions are measured by Regional Price Parities, a division-level CPI measure, and the real minimum wage. A medical-debt indicator serves as a background measure of household balance-sheet stress; it equals one when a state’s medical-debt burden exceeds a predefined threshold and zero otherwise, which explains the mass at zero in Table 1.

All variables are aligned at the weekly frequency. Policy and fiscal variables that occur on discrete dates are coded to match the weekly panel. Lagged terms are constructed so that the information set at week t uses only variables observed at or before $t-1$. Continuous covariates are standardized to mean zero and unit variance immediately prior to estimation, after the summary statistics reported in Table 1 are computed; the table therefore reports each variable in its original units, not in standardized form. Standardization ensures coefficients are comparable across regressors and sectors. Binary indicators enter in their raw 0/1 form and are not standardized.

6.4 Spatial weights

Spatial linkages across states are summarized by a predetermined $N \times N$ weights matrix W . In the baseline specification, W is a row-normalized four-nearest-neighbor matrix based on state centroid distances: $W_{ik} > 0$ if state k is among the four nearest neighbors of state i by centroid distance, and $W_{ik} = 0$ otherwise. Each row sums to one by construction. The matrix

is held fixed throughout estimation and is used solely to model spatial clustering in exposure to the nationwide latent factor through the SAR loading structure in equation (6). The four-nearest-neighbor specification is preferred over contiguity because Alaska and Hawaii share no land borders with other states and would be isolated units under a contiguity-based matrix. Assigning neighbors by centroid distance ensures that all fifty states participate in the spatial structure.

6.5 Summary statistics

Table 1 reports summary statistics for the dependent variables and key covariates used in the baseline estimation. The spending series show substantial heterogeneity across sectors: Transportation has a negative mean level relative to baseline over the sample period, consistent with its negative factor loading, while General Merchandise and Home Improvement show the largest positive means, consistent with the goods consumption boom during the recovery. The maximum value for Home Improvement reflects a genuine but extreme state-week observation during the early pandemic period and is retained in the baseline sample as part of the reported spending series. EIP variables are reported in their scaled units after division by 10,000 to bring them into a comparable range with the other covariates. The dataset is compiled from the Opportunity Insights Economic Tracker and complementary sources including the U.S. Census Bureau and the BLS.

7 Empirical Results

This section interprets posterior results from the spatial dynamic factor model estimated on the weekly state-by-sector panel. The model decomposes sectoral spending, relative to baseline, into three components: a persistent latent national factor f_t , heterogeneous state-by-sector exposure to that factor through the loading vectors A_i with spatial structure, and within-sector dynamics captured by the VAR(1) matrix Γ . Statistical significance throughout

Table 1: Summary statistics

Variable	N	Mean	Std. Dev.	Min	Max
<i>Dependent variables (spending relative to January 2020 baseline)</i>					
Apparel & Accessories	11,500	0.054	0.185	-0.726	0.863
Accommodations & Food Services	11,500	0.030	0.209	-0.756	0.522
Arts, Entertainment & Recreation	11,500	0.113	0.438	-0.873	2.970
Auto & Gas	11,500	0.241	0.197	-0.465	0.898
Health Care	11,500	0.065	0.220	-0.772	1.740
Home Improvement	11,500	0.296	0.262	-0.469	7.446
General Merchandise	11,500	0.366	0.250	-0.334	1.420
Grocery & Food	11,500	0.140	0.107	-0.261	1.550
Sport, Hobby, Books & Music	11,500	0.303	0.270	-0.809	3.872
Transportation	11,500	-0.115	0.351	-0.924	2.870
<i>Covariates</i>					
Vaccination progress	11,500	0.039	0.101	0.000	2.331
Initial unemployment claims	11,500	0.304	0.553	0.009	7.990
Nonessential business closure (indicator)	11,500	0.002	0.046	0	1
Stay-at-home start (indicator)	11,500	0.003	0.051	0	1
Employment growth	11,500	-0.041	0.098	-0.363	0.707
Regional Price Parity (RPP)	11,500	100.8	3.987	89.6	111.5
Division-level CPI	11,500	115.9	8.355	101.7	131.8
Real minimum wage (\$)	11,500	11.07	2.842	7.235	17.21
Medical debt indicator	11,500	0.006	0.078	0	1
EIP1 (\$ billions)	11,500	5.125	5.695	0.000	29.60
EIP2 (\$ billions)	11,500	2.196	2.886	0.000	15.41
EIP3 (\$ billions)	11,500	5.912	8.235	0.000	44.77
Lagged personal income (\$000s)	11,500	432.5	529.6	35.0	3,327.9

Notes: The panel covers 50 U.S. states over 230 effective estimation weeks used by the VAR(1) component. Spending variables are the seasonally adjusted level of spending relative to its January 2020 baseline, following the Opportunity Insights spending index. Summary statistics for continuous covariates are reported in each variable's original units; these covariates are standardized to mean zero and unit variance immediately prior to estimation, after the statistics reported here are computed. Binary indicators are not standardized. EIP variables are reported in billions of dollars at the state level. Lagged personal income is reported in thousands of dollars. The maximum value for Home Improvement reflects a genuine but extreme state-week observation during the early pandemic period and is retained in the baseline sample.

refers to whether the 95% equal-tailed credible interval excludes zero. Because continuous covariates enter the model in standardized form, regression coefficients are interpreted as the change in weekly spending, relative to baseline, associated with a one-standard-deviation increase in the regressor.

Figure 2 provides an initial illustration of the geographic and sectoral heterogeneity that motivates the analysis. The two panels show spending, relative to baseline, across states and sectors at the collapse week and at the initial rebound, plotted on a common color scale to facilitate comparison. The maps make clear that the same nationwide disruption generated markedly different local outcomes across states and categories, both at the trough and during the recovery.

7.1 Factor loadings and geographic exposure

Table 2 summarizes the posterior distribution of state-level factor loadings by sector. Excluding the normalization loading (Alabama’s Apparel and Accessories loading, fixed to one and not an estimated posterior parameter), 95.6% of the remaining 499 state-sector loadings have 95% credible intervals excluding zero, confirming that the estimated national factor reaches nearly all states and sectors. At the same time, the magnitude and direction of exposure vary substantially, and this heterogeneity is one of the paper’s main empirical findings.

Mean exposure differs sharply across categories. Sport, Hobby, Books, and Music exhibits the largest mean loading (3.16), followed by General Merchandise (2.42), Auto and Gas (2.13), and Apparel and Accessories (1.69), with essentially all state-level loadings significant in these sectors. At the other extreme, Arts, Entertainment, and Recreation (−1.79) and Transportation (−1.93) display negative mean loadings, implying that spending in these categories tended to move against the direction of the national factor for most states. Contact-intensive activities remained partly decoupled from the broader recovery that lifted goods-oriented consumption. This pattern is consistent with the structural evidence in Miescu and Rossi (2021), who find that these two categories recorded the largest

Table 2: Factor-loading summaries by sector

Sector	Mean(A)	SD(A)	Share $ A \leq 0.10$	Share significant
Apparel & Accessories	1.693	0.311	0.0%	100.0%
Accommodations & Food	0.857	0.230	0.0%	96.0%
Arts, Entertainment & Recreation	-1.793	1.291	0.0%	96.0%
Auto & Gas	2.134	0.286	0.0%	100.0%
Health Care	-0.088	0.594	10.0%	66.0%
Home Improvement	1.396	0.554	0.0%	100.0%
General Merchandise	2.418	0.418	0.0%	100.0%
Grocery & Food	1.051	0.319	0.0%	100.0%
Sport, Hobby, Books & Music	3.159	0.635	0.0%	100.0%
Transportation	-1.929	0.846	0.0%	98.0%

Notes: Posterior summaries are reported for state-level factor loadings by sector. “Share significant” denotes the percentage of state loadings in that sector whose 95% equal-tailed credible interval excludes zero.

expenditure contractions in response to a common COVID-19 uncertainty force, confirming that both approaches capture the same underlying national driver. The positive but modest loading on Grocery and Food mirrors the finding in Miescu and Rossi (2021) that grocery spending is the only expenditure category to increase in response to the same force, consistent with necessity-driven consumption operating through a fundamentally different channel.

Beyond sector means, the cross-state dispersion in loadings reveals important geographic structure. General Merchandise has a standard deviation of only 0.42 relative to a mean of 2.42, suggesting relatively uniform national exposure. Arts, Entertainment, and Recreation has a standard deviation of 1.29, indicating sharply divergent state-level responses. State-level minimum and maximum loadings within each sector are reported in Table D1 of Appendix D of the Online Appendix.

Health Care is the clearest exception to the general pattern. Its mean loading is near zero (-0.09), its standard deviation is 0.59, and only 66% of state loadings are significant, by far the lowest share of any sector. Despite being one of the sectors most directly implicated by the pandemic, Health Care is least organized by a common national force. Local institutional

features such as provider structure, insurance arrangements, and state-specific health-care systems appear to dominate, generating idiosyncratic rather than geographically clustered exposure.

Figure 3 provides a visual summary of these patterns, plotting mean loadings and cross-state dispersion by sector alongside the spatial dependence estimates discussed below.

7.2 Spatial clustering in factor exposure

Table 3 reports the posterior distribution of the spatial dependence parameter ρ_j for each sector. In most sectors the posterior mean is large and the 95% credible interval excludes zero, confirming that neighboring states tend to share similar sensitivity to the national factor. Sport, Hobby, Books, and Music ($\rho = 0.95$), General Merchandise ($\rho = 0.92$), and Auto and Gas ($\rho = 0.92$) exhibit the strongest spatial clustering. Health Care ($\rho = 0.38$) is the only sector where spatial dependence is substantially weaker and the credible interval covers values near zero, consistent with its idiosyncratic loading pattern documented above.

Table 3: Spatial dependence parameters ρ_j

Sector	Mean	CI _L	CI _U
Apparel & Accessories	0.897	0.817	0.964
Accommodations & Food	0.772	0.612	0.907
Arts, Entertainment & Recreation	0.821	0.730	0.901
Auto & Gas	0.915	0.851	0.970
Health Care	0.378	0.039	0.677
Home Improvement	0.866	0.766	0.948
General Merchandise	0.922	0.864	0.972
Grocery & Food	0.818	0.687	0.926
Sport, Hobby, Books & Music	0.951	0.909	0.986
Transportation	0.886	0.812	0.950

Notes: Posterior means and 95% equal-tailed credible intervals for the sector-specific spatial dependence parameters ρ_j .

The large spatial dependence estimates imply that nationwide forces do not transmit uniformly across geography even within a sector. States that share economic corridors or

industrial structures respond more similarly to the same national movement, generating regional clusters of amplification or attenuation. This geographic organization of factor sensitivity means that national aggregates can mask persistent regional divergence even when the underlying national force is common to all states.

7.3 Dynamic propagation and the role of the common factor

Table 4 reports the diagonal elements of Γ , which capture own-sector persistence in weekly spending relative to baseline. As noted in Section 4, part of this persistence is mechanical given the cumulative construction of y_{ijt} , so the magnitudes below should not be read as pure measures of behavioral habit formation. All ten diagonal elements are positive and significant, indicating substantial inertia within each category. None of the off-diagonal elements are significant at the 95% level, as noted in the table. Cross-sector co-movement is therefore captured by the common national factor and spatially clustered exposure rather than by direct lagged transmission across categories.

Table 4: Own-lag persistence: diagonal elements of Γ

Sector	Mean	CI _L	CI _U	Sig.
Apparel & Accessories	0.606	0.473	0.740	Yes
Accommodations & Food	0.648	0.489	0.807	Yes
Arts, Entertainment & Recreation	0.413	0.337	0.488	Yes
Auto & Gas	0.562	0.395	0.729	Yes
Health Care	0.419	0.299	0.540	Yes
Home Improvement	0.533	0.454	0.611	Yes
General Merchandise	0.550	0.412	0.689	Yes
Grocery & Food	0.488	0.295	0.681	Yes
Sport, Hobby, Books & Music	0.422	0.348	0.497	Yes
Transportation	0.486	0.401	0.572	Yes

Notes: Posterior means and 95% equal-tailed credible intervals for the diagonal elements of Γ . No off-diagonal element of Γ is significant at the 95% level.

This result connects to the sectoral pattern already noted in Section 7: the categories that Miescu and Rossi (2021) identify as suffering the largest expenditure contractions in

their structural VAR are the same sectors whose spending co-moves through the latent factor here rather than through direct lagged transmission, reinforcing that the two approaches are capturing the same underlying national force.

Table 5 reports the posterior distribution of the factor AR(1) coefficient ϕ and innovation variance σ_u^2 . The posterior mean of ϕ is 0.98 with a tight credible interval, confirming that the national factor is highly persistent and moves slowly relative to the weekly sampling frequency. As discussed in Section 4, part of this persistence is mechanical given the cumulative construction of y_{ijt} , so this magnitude should not be read purely as a measure of how slowly underlying economic conditions evolve. Because the prior on ϕ is truncated to the stationary region $|\phi| < 1$, posterior inference is conducted entirely over stationary AR(1) processes. The posterior for ϕ is therefore well-defined even though the posterior mass is concentrated close to the upper boundary, with a posterior mean of 0.980 and a 95% credible interval of [0.958, 0.997]. This Bayesian treatment avoids relying on conventional Gaussian or asymptotic approximations for the sampling distribution of an autoregressive coefficient near the unit-root boundary, where classical inference can be nonstandard (Sims and Uhlig, 1991). The small innovation variance is nonetheless consistent with a slow-moving common force rather than high-frequency noise. Figure 4 plots the estimated factor path over the full sample period, showing a brief dip below zero early in 2020 followed by a sustained upward drift for the remainder of the sample, with the factor reaching its highest estimated level toward the end of the period in 2024.

Table 5: Latent factor dynamics: persistence and innovation variance

Parameter	Mean	CI _L	CI _U
ϕ	0.980	0.958	0.997
σ_u^2	0.607×10^{-3}	0.342×10^{-3}	0.946×10^{-3}

Notes: Posterior means and 95% equal-tailed credible intervals for the AR(1) persistence parameter ϕ and the factor innovation variance σ_u^2 .

7.4 Sector-specific covariate effects

Figure 5 provides a compact overview of how the main covariates relate to spending, relative to baseline, across sectors. Two broad patterns stand out. Labor-market stress and reopening variables matter unevenly across categories, while price conditions and fiscal transfers operate through distinct sector-specific channels. Full posterior coefficient tables for each sector are reported in Appendix D.2 of the Online Appendix.

Home Improvement displays one of the clearest sector-specific patterns: spending, relative to baseline, is positively associated with the first round of Economic Impact Payments and with regional price parity, while it is negatively associated with CPI inflation. The response was therefore strongest where stimulus support and local purchasing-power conditions were more favorable, consistent with the pandemic-era reallocation toward home-based consumption (Alexander and Karger, 2023; Dao et al., 2024).

Reopening-sensitive categories display a related but distinct pattern. In Accommodations and Food Services and Arts, Entertainment and Recreation, vaccination progress is positively and significantly associated with spending relative to baseline, consistent with stronger recovery where public-health conditions improved and in-person activity normalized. Transportation responds most clearly to closure and stay-at-home timing variables and to broader labor-market recovery, with employment entering strongly and positively. Normalization therefore did not operate through a single channel: different reopening-sensitive sectors responded to different parts of the public-health and economic recovery process.

Initial unemployment claims are negatively and significantly associated with spending relative to baseline across a wide range of discretionary sectors, including Apparel, Accommodations and Food Services, Arts, Entertainment and Recreation, Auto and Gas, Health Care, Sport, Hobby, Books, and Music, and Transportation. Grocery and Food provides an informative contrast: initial claims enter positively and significantly there, consistent with substitution toward at-home consumption during periods of economic stress (Guerrieri et al., 2022). This essential-versus-discretionary contrast reinforces the broader finding that

common forces generated uneven local outcomes because sectors responded through different spending margins.

Fiscal-transfer coefficients vary substantially across rounds and sectors and should be interpreted with care. Because the dependent variable y_{ijt} measures the level of spending relative to its January 2020 baseline rather than a growth rate, a negative coefficient on an EIP variable does not imply that transfers caused an outright reduction in spending. It indicates that spending, relative to baseline, ran lower during disbursement weeks than the other covariates and the common factor alone would predict, conditional on the other controls. EIP timing also frequently overlaps with changes in non-pharmaceutical interventions and reopening phases, so these estimates are best interpreted as reduced-form timing associations rather than structural spending elasticities.

8 Robustness checks

We assess the robustness of the main specification along two dimensions, removing one dynamic channel of co-movement at a time. The first exercise re-estimates the model with the sectoral VAR component switched off ($\Gamma = 0$), isolating the common factor and observed covariates as the remaining sources of co-movement. The second does the reverse: it re-estimates the model with the common factor switched off ($A_i f_t = 0$) while allowing the full unrestricted $J \times J$ VAR matrix Γ to capture within- and cross-sector lagged dynamics. Together these two exercises clarify whether the estimated national factor is merely absorbing sectoral VAR dynamics, and whether apparent cross-sector spillovers emerge when the latent national factor is removed.

8.1 Removing sectoral VAR dynamics

We re-estimate the model imposing $\Gamma = 0$, removing own-sector persistence and cross-sector lagged propagation so that remaining common movement is captured solely by observed

covariates and the spatial latent factor.

The no-VAR results are consistent with the main specification. Table D14 in Appendix D.3 of the Online Appendix reports posterior summaries for the latent factor process under this specification. The latent national factor remains highly persistent, with a posterior mean of $\phi = 0.9967$ and a 95% credible interval of $[0.9923, 0.9989]$. The factor innovation variance is small, with a posterior mean of 0.00082 and a 95% credible interval of $[0.00065, 0.00102]$. The spatial exposure channel also remains strong: 94.8% of state-sector loadings have credible intervals excluding zero, comparable to the 95.6% share under the main specification (Section 7.1). Spatial dependence is positive across all sectors, with especially high posterior means for Sport, Hobby, Books, and Music; Apparel and Accessories; Auto and Gas; General Merchandise; and Home Improvement.

Figure 6 compares the posterior mean of the latent national factor from the full model and the no-VAR specification. Because the latent factor and factor loadings are jointly identified only up to a normalization, the two paths are not directly comparable in magnitude. The relevant comparison is their temporal evolution. Both specifications recover a persistent national factor with a similar time pattern, indicating that the estimated common national component is not an artifact of the sectoral VAR structure.

Even when sectoral dynamic propagation is removed, the data continue to exhibit a persistent national factor with geographically clustered sector-specific exposure, supporting the main conclusion of the paper.

8.2 Removing the common factor

We re-estimate the model imposing $A_i f_t = 0$ for all i , removing the shared national component entirely and retaining only the full unrestricted $J \times J$ VAR matrix Γ to capture lagged cross-sector dynamics. For this specification we use equal prior variances for diagonal and off-diagonal elements of Γ , unlike the asymmetric prior described in Section 5.3, so that the no-factor results do not mechanically shrink cross-sector spillovers toward zero. This is the

natural complement to the no-VAR exercise: rather than asking whether the factor survives without Γ , we ask how estimated VAR dynamics change when the common factor is removed.

The no-factor results show that the sectoral VAR absorbs substantially more dynamic dependence when the latent national factor is excluded. Table D12 in Appendix D.3 of the Online Appendix reports the diagonal elements of Γ under this specification. Own-sector persistence remains positive and statistically significant in most sectors, confirming strong within-sector dynamics even without the latent factor.

More importantly, Table D13 in Appendix D.3 of the Online Appendix reports the statistically significant off-diagonal elements of Γ under this specification. In contrast to the full model, several cross-sector coefficients become significant when the common factor is removed. Lagged Accommodations and Food Services spending, relative to baseline, positively predicts Arts, Entertainment and Recreation and Transportation, while lagged Home Improvement spending positively predicts Transportation. Lagged Accommodations and Food Services spending negatively predicts Sport, Hobby, Books, and Music, and lagged Health Care spending positively predicts Sport, Hobby, Books, and Music.

The disappearance of these cross-sector spillovers once the latent factor is included reflects a form of omitted-variable bias in the VAR-only specification. Sectors such as Accommodations and Food Services, Arts and Entertainment, and Transportation all load on the same national factor, which captures the common trajectory of contact-intensive activity during the pandemic and recovery. The sectoral composition here echoes the pattern already noted in Section 7: these are the same categories that Miescu and Rossi (2021) identify as bearing the largest expenditure contractions in their structural VAR. In the absence of the shared factor, the VAR must account for their co-movement using lagged values of correlated sectors. The significant off-diagonal elements in the no-factor specification therefore do not appear to represent direct cross-sector transmission from one sector to another; they are consistent with the autocorrelation of a common omitted force. Once f_t is included and the spatially structured loadings A_i absorb this common variation, the spurious cross-sector lags

lose their explanatory power and become indistinguishable from zero.

9 Conclusion

This paper studies why U.S. consumer spending evolved so differently across states and sectors during the pandemic and recovery period from 2020 through mid-2024. Using a weekly state-by-sector panel, we estimate a Bayesian spatial dynamic factor model in which sectoral spending, relative to its January 2020 baseline, is driven by observed state-week fundamentals and a latent national factor, with geographic dependence entering through spatially structured sector-specific loading vectors.

Four findings stand out. The latent national factor is highly persistent, indicating that a slow-moving common component is central for explaining post-2020 spending dynamics rather than transitory noise. Exposure to that national component is strongly spatially clustered in most sectors: neighboring states share similar sensitivity to nationwide forces, so the same aggregate movement generates regional clusters of amplification or attenuation rather than uniform local responses. Sector-specific mechanisms differ sharply across categories. Home improvement spending, relative to baseline, is positively associated with the first round of Economic Impact Payments and with local cost-of-living conditions, consistent with housing-related reallocation amplified by stimulus and purchasing power. Reopening-sensitive sectors such as accommodations, arts and recreation, and transportation are more tightly linked to vaccination progress, labor-market conditions, and closure timing, indicating that normalization operated through several distinct channels rather than a single reopening effect. Finally, once common dynamics and spatial exposure are accounted for, spending persistence is largely within-sector: own-lag coefficients are strong but lagged cross-sector spillovers are not credibly different from zero. This last result has a direct methodological implication: studies that rely on sectoral VARs without controlling for a common aggregate force will attribute to inter-sectoral transmission what is in fact a shared national component.

The policy implication of these findings is concrete. National programs such as fiscal transfers or reopening measures are uniform in design but their associations with local spending are not uniform in practice. Because exposure to common forces is spatially clustered, the same federal policy can generate regionally concentrated benefits or costs depending on a state’s economic structure and sector composition. Evaluating broad-based stabilization policies through national aggregates alone therefore obscures economically meaningful local heterogeneity that high-frequency state-by-sector monitoring can identify in real time, even when national indicators suggest recovery is on track.

Conflict of interest statement

The authors declare that they have no conflict of interest.

Data and code availability

The spending data used in this paper are derived from the Opportunity Insights Economic Tracker and publicly available complementary sources. Replication code and processed analysis files will be made available through the JAE Data Archive upon acceptance, subject to the data-use restrictions of the underlying sources.

Supporting information

Additional supporting information, including theoretical derivations, Monte Carlo evidence, the full Gibbs sampler and conditional posteriors, and additional empirical results, may be found in the Online Appendix accompanying this article.

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Figures

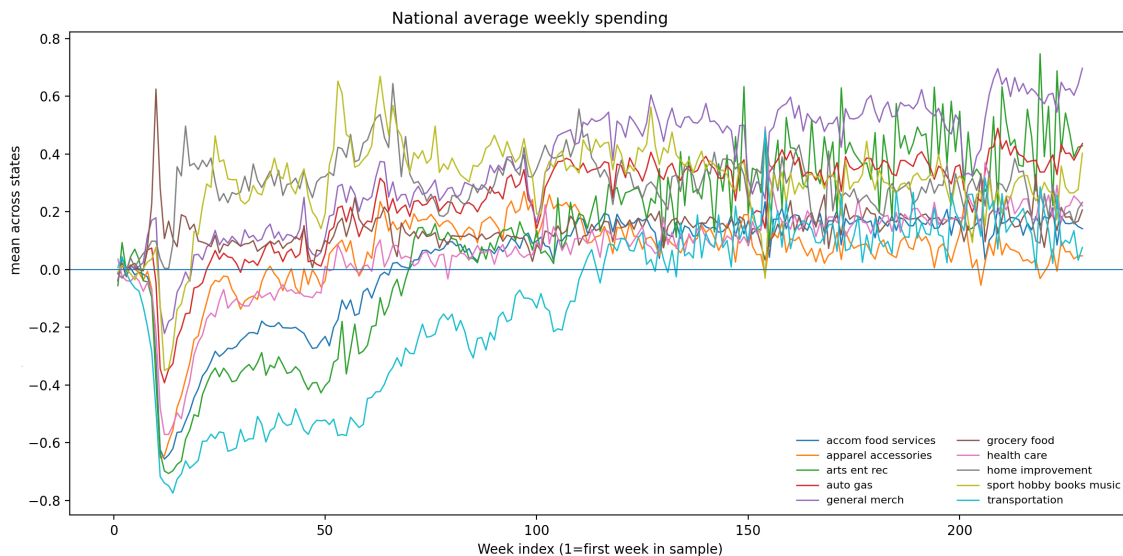
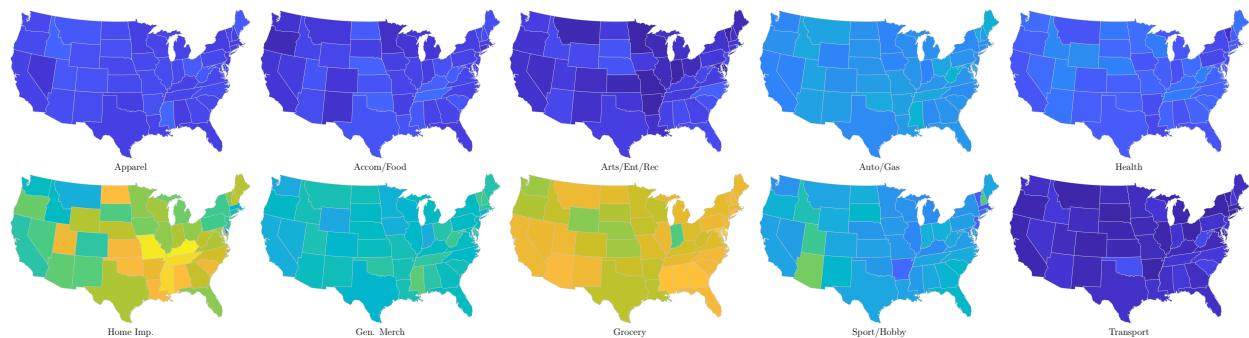
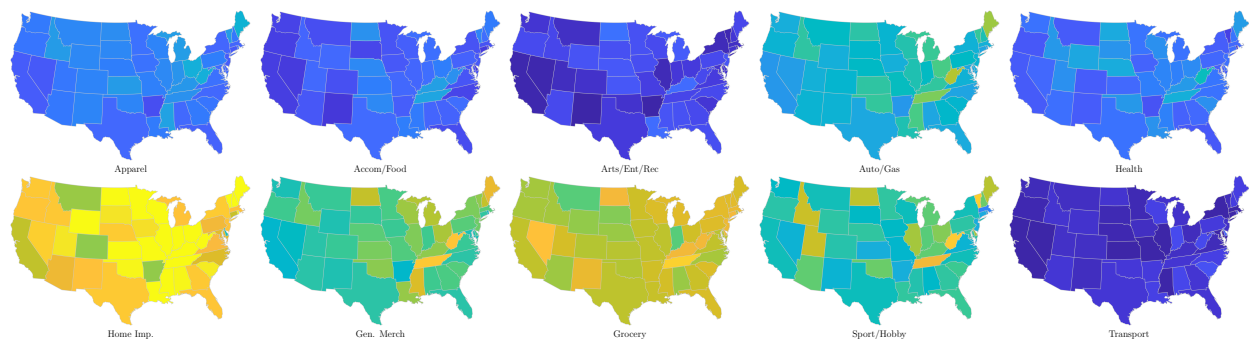


Figure 1: National average weekly spending relative to its January 2020 baseline by sector, January 2020 through June 2024. Each series plots the mean level, relative to baseline, across the fifty states. The divergence between contact-intensive sectors such as Transportation and Arts and Entertainment and goods-oriented sectors such as General Merchandise and Home Improvement is the primary empirical motivation for the spatial dynamic factor model estimated in the paper.



(a) Collapse week (week 12).



(b) Initial rebound period (week 15).

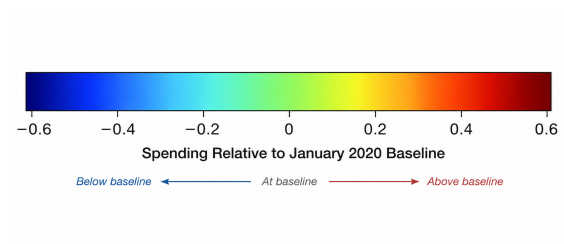


Figure 2: Geographic and sectoral heterogeneity during the collapse and initial rebound. Panel (a) reports week 12 (collapse), while panel (b) reports week 15 (initial rebound). All maps use a common color scale. Blue indicates spending further below its January 2020 baseline, and red indicates spending further above baseline, allowing direct comparison of the magnitude and spatial distribution of consumer spending across states and sectors.

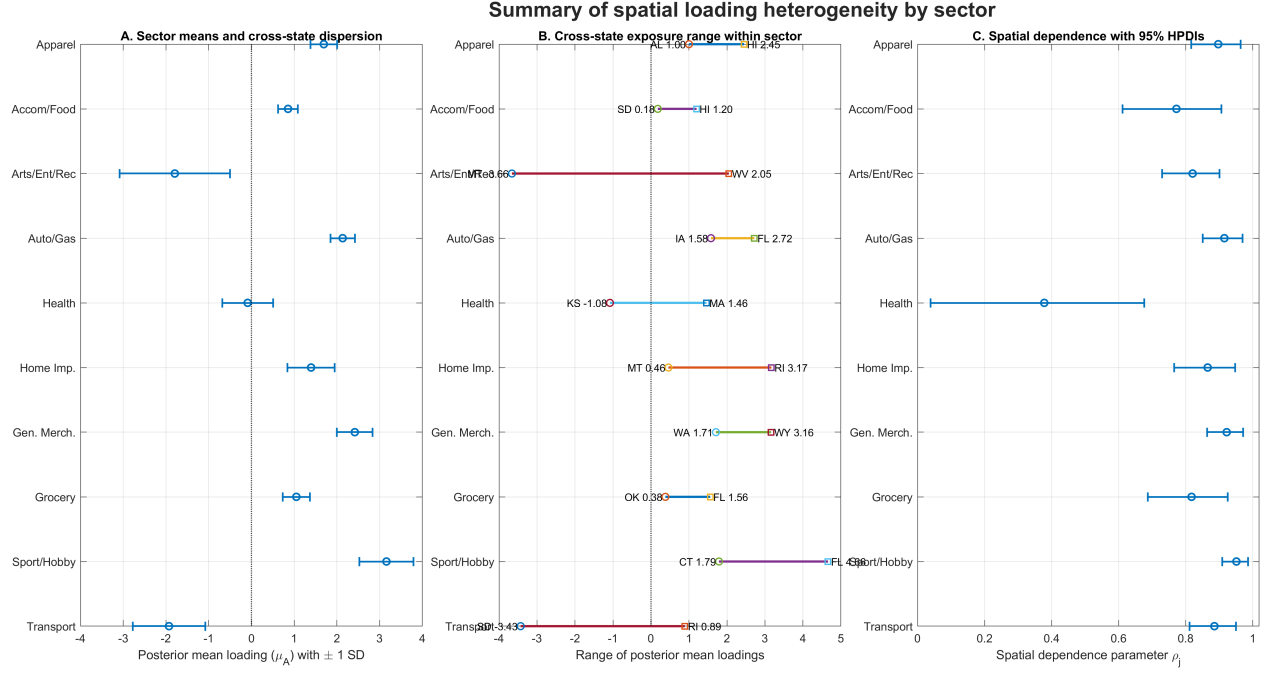


Figure 3: Summary of spatial loading heterogeneity by sector. Panel (a) reports sector-specific posterior mean loadings and cross-state dispersion. Panel (b) reports the minimum and maximum posterior mean loading in each sector, with the corresponding states labeled. Panel (c) reports posterior means and 95% equal-tailed credible intervals for the spatial dependence parameter ρ_j .

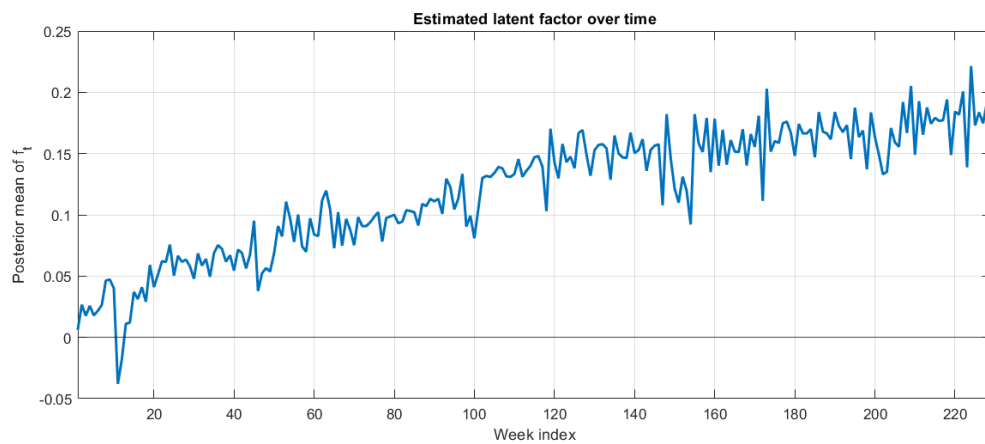


Figure 4: Estimated latent factor over time. The figure plots the posterior mean of the common national factor f_t over the sample period, summarizing the evolution of the latent national component common across states and sectors.

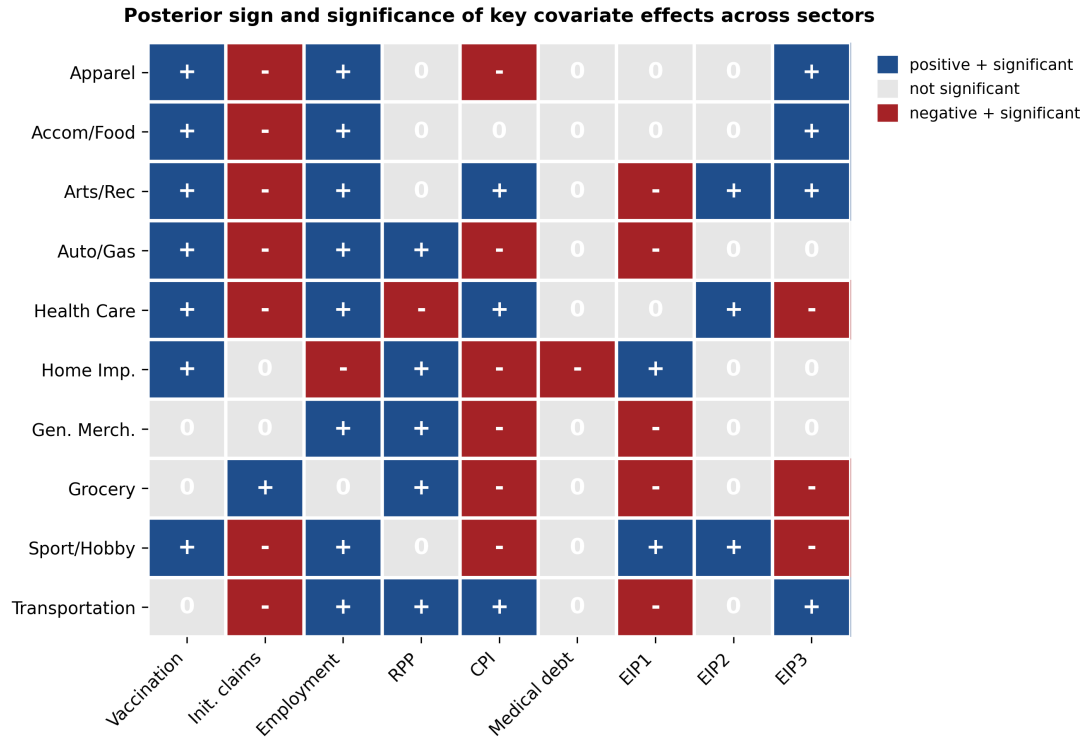


Figure 5: Posterior sign and significance of key covariate effects across sectors. A “+” denotes a positive coefficient whose 95% credible interval excludes zero, a “-” denotes a negative coefficient whose interval excludes zero, and “0” denotes an interval that includes zero.

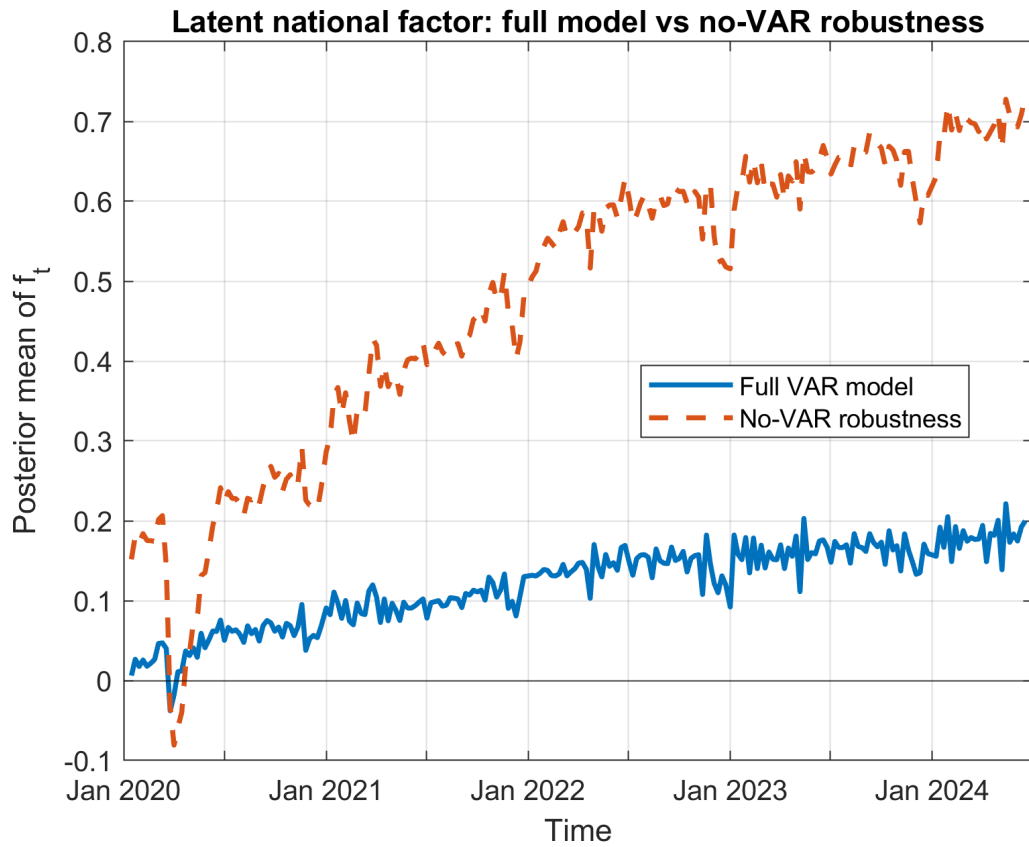


Figure 6: Posterior mean of the latent national factor under the full model and the no-VAR robustness specification. The magnitudes differ because of factor normalization, but both specifications recover a similar temporal evolution of the national component.